geneticist Ronald Fisher wrote *The Genetical Theory of Natural Selection*, which combined Darwin's theory of natural selection with the ideas of heredity that the 19th-century Austrian scientist Gregor Mendel had developed. In 1937, Ukrainian–American geneticist Theodosius Dobzhansky put forward the idea that regularly occurring genetic mutations are sufficient to provide the genetic diversity—and therefore different traits—that makes natural selection possible. He wrote that evolution was a change in the frequency of an "allele" in the gene pool, an allele being one of the alternative forms of a gene that arise by mutation.

A mutation is a permanent alteration in the sequence of deoxyribonucleic acid (DNA), the molecule that makes up a gene in one individual, resulting in a sequence that differs from that of other members of the species. Mutations may occur as the result of the miscopying of DNA during cell division, or they may be caused by environmental factors, such as damage resulting from the sun's ultraviolet radiation. One mutation might affect only the individual organism carrying it, whereas another might affect all its offspring and future generations.

Inherited mutations may or may not alter an individual's phenotype – its physical traits and behavior. If mutations do affect the phenotype, they may be to its advantage or disadvantage, helping or hindering an organism's ability to survive and reproduce successfully. If they hinder, they are likely to disappear from the population; if they help an organism adapt better to its environment, they become more common over the course of generations. Over time, they may produce large enough divergences from the parent population for a new species to evolve—a process called speciation.

Mutation rates are usually very low, but the process is ever-present. The changes may be beneficial, neutral, or harmful. They do not occur in response to an organism's needs, and are, in that respect, random. However, some types of mutations occur more frequently than others. Scientists now